

# Session Objectives

- **Review the pathophysiology of angina pectoris**
- **Distinguish between the types of anginas and the appropriate therapeutics.**
- **Describe the mechanism of nitrates in angina. Include adverse effects and specific drug interactions, mainly drugs for erectile dysfunction**
- **Review the mechanism of beta-adrenergic receptor antagonists. Identify patients which this class of drug is contraindicated.**
- **Describe the differences between the classes of calcium channel blockers.**
- **Identify other drugs used to manage angina, including ranolazine.**
- **Review the mechanisms of antiplatelet drugs.**

# Nitric oxide (NO) donors

- Nitroglycerin is denitrated by glutathione-S-transferase.
- Free nitrite is converted to NO.
- Another metabolic release of NO is by mitochondrial aldehyde reductase.
- Tolerance: occurs with continuous use. May be due to depletion of the sulfur group or inhibition of aldehyde reductase. An 8-hour drug holiday is required to restore action.
- NO donor results: Increased venous capacitance, decreased preload, reduced venous return = less cardiac output

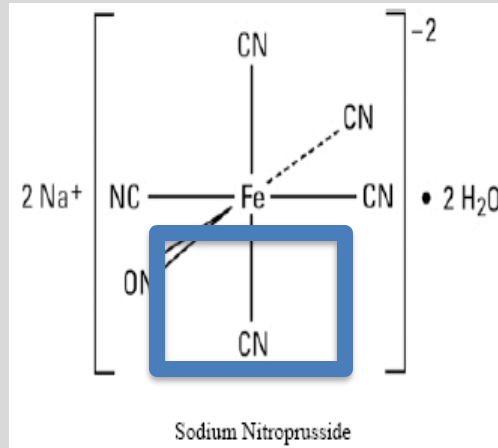
# Nitrates: Clinical Uses

- Immediate relief of angina: sublingual nitroglycerin
- Severe angina: IV nitroglycerin
- Recurrent typical and atypical angina: oral, buccal, transdermal
- Other uses: Heart Failure (decreases preload)  
Unresponsive hypertension (sodium nitroprusside has added arteriolar vasodilator action).

# Nitrates: Adverse Effects

- Throbbing, vascular headache
- Vasodilation, hypotension (orthostatic) and compensatory tachycardia, with increased ventricular contractility
- Compensatory retention of salt and water due to enhanced renin-angiotensin II-aldosterone.
- NO activates guanylyl cyclase. Increased cGMP decreases platelet activation. Use as antiplatelet not established.
- Drug interactions: PDE5 inhibitors: sildenafil, vardenafil, tadalafil.

# Sodium Nitroprusside



NEW YORK INSTITUTE  
OF TECHNOLOGY

College of Osteopathic  
Medicine

- Short-acting NO donor, IV use for **hypertensive emergencies**. Dilates arterioles and venules. Decreases afterload and preload.
- Adverse effect: Rare accumulation of cyanide. Drug is a complex of iron, cyanide, and nitroso moiety.
- Antidote: sodium thiosulfate or hydroxycobalamin (vitamin B<sub>12</sub>).

# Calcium Channel Blockers (CCB)

- **Arterioles relax, TPR and afterload decrease.**  
Blood pressure falls, heart works less for systole. Common to all CCBs for antianginal and antihypertensive use.
- **Verapamil and diltiazem: Heart automaticity and ventricular contractility are decreased.**  
Class IV antiarrhythmic, antihypertensive, and antianginal use.

# Verapamil and Diltiazem (non-DHP)

NEW YORK INSTITUTE  
OF TECHNOLOGY

College of Osteopathic  
Medicine

- Arteriole and Heart action; Verapamil > Diltiazem
- Resist the SANS reflex, replaces beta blockers in COPD, asthma, diabetes patients
- Uses: hypertension, angina, Class IV antiarrhythmic (manages SVT, atrial fibrillation)
- Adverse Effects: Hypotension without high HR. SA and AV node depression. Negative inotropic effect = contraindicated in CHF.

# CCB Adverse Effects

- Dizziness, headache, edema
- Tachycardia (DHP)
- Bradycardia (non-DHP)
- Constipation (most with verapamil)
- Gingival overgrowth (review past drugs??)
- Grapefruit drug interactions (CYP 3A4)
- Verapamil increases digoxin plasma levels.
- Antidote: Supportive care; IV calcium, insulin



# Beta-adrenergic receptor antagonists

- Beta 1: decrease HR, CO, and renin from JXA
- Use: hypertension, MI, CHF, antianginal (except vasospastic), Phase II antiarrhythmic
- Beta 1: (atenolol, metoprolol)—AV block most adverse effect
- Beta 1, 2: (nadolol, propranolol, timolol). Bronchoconstriction, hypoglycemia, complicates PVD (Beta 2)
- Labetalol and Carvedilol (alpha and beta blocker)
- Pindolol and Acebutolol (ISA)-can worsen angina
- Esmolol (ultrashort acting)
- Nebivolol (nitric oxide effect)
- Antidote: supportive therapy, glucagon

# Ranolazine

- Reduces late sodium current. Prevents calcium entry via the  $\text{Na}^+/\text{Ca}^{2+}$  exchanger.
- Angina prophylaxis (chronic)
- Adverse effects: nausea, constipation, dizziness. CYP inhibitors increase ranolazine action. Prolongs QT interval.

# Antiplatelet drugs

- Aspirin **irreversibly** blocks COX 1,2 (NASIDs reversible). Analgesic, anti-inflammatory, antipyretic
- ADP receptor blockers (clopidogrel)
- Glycoprotein IIb/IIIa inhibitors (abciximab)
- Phosphodiesterase inhibitors (cilostazol, dipyridamole)—for peripheral artery disease

# Aspirin

- Prevents (primary) cardiovascular disease and recurrence (secondary)
- Consider individual patient (benefit vs risk)
- Adverse effects: GI bleed, hypersensitivity, metabolic acidosis (overdose). Uncoupling of oxidative phosphorylation.
- Salicylates not for pediatric patients.

# ADP receptor blockers

- Ticlopidine, clopidogrel, prasugrel, ticagrelor, and cangrelor
- All are prodrugs, but ticagrelor and **cangrelor (IV)**.
- Clopidogrel is activated by 2C19 (mutations or drug inhibiting 2C19 will reduce antiplatelet effect).
- Clopidogrel: for unstable angina or non-ST-elevation acute myocardial infarction (NSTEMI) in combination with aspirin (also STEMI, recent MI, and PAD)
- All prevent thrombosis: Considered standard practice in patients undergoing stenting. And for acute coronary syndromes.
- Adverse effect: bleeding

# GP IIb/IIIa Inhibitors

NEW YORK INSTITUTE  
OF TECHNOLOGY

College of Osteopathic  
Medicine

- **Abciximab (monoclonal antibody)-No longer available in U.S.**
- **Reversible inhibitors: Eptifibatide, tirofiban. Short half lives= administer by continuous infusion.**
- **Use: Acute coronary syndrome or PCI in those with higher thrombus risk, unable to tolerate ADP receptor blockers, or aspirin allergy.**
- **Adverse effect: bleeding**
- **Pathology Correlate: Glanzmann disease results from absent or defective GP IIb/IIIa.**